# Hyperbolic Functions: Integration

Questions 1–25

**Question 1.** Find:

a) $\int \sinh x \, dx$

b) $\int \cosh x \, dx$

c) $\int \operatorname{sech}^2 x \, dx$

d) $\int \sinh 3x \, dx$

e) $\int \cosh \frac{x}{2} \, dx$

f) $\int \operatorname{sech}^2 9x \, dx$

**Question 2.** Evaluate the following definite integral:

$$\int_{\ln 2}^{\ln 3} \sinh x \, dx$$

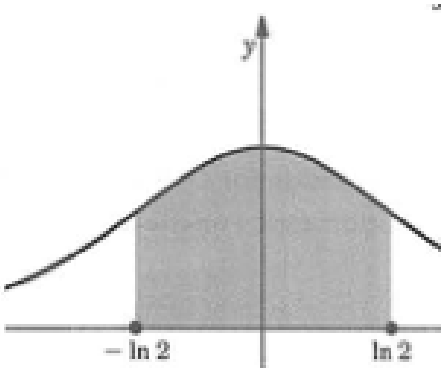
**Question 3.** Evaluate the following definite integral:

$$\int_{\ln 3}^{\ln 4} \cosh 2x \, dx$$

**Question 4.** Evaluate the following definite integral:

$$\int_{0}^{\ln 2} (15 \cosh 3x - 9 \sinh 3x) \, dx$$

**Question 5.** The diagram below shows the curve $y = \text{sech}^2 x$. Show that the exact shaded area is $\frac{6}{5}$.




**Question 6.** Find the exact value of:

$$\lim_{\delta x \to 0} \sum_{x=0}^{\ln 2} (8\cosh 4x - 5\sinh 3x)\,\delta x$$

**Question 7.** It is given that $f(x) = 6\sinh 3x + 15\cosh 3x$. Find the exact positive value of $k$ for which the region between the curve $y = f(x)$, the $x$-axis, the $y$-axis, and the line $x = k$ is 8. **Prerequisite knowledge for Q8: Integration of Reciprocal Trigonometric Functions**

**Question 8.** Show that

$$\int_{-\pi/4}^{\pi/4} (\sec^2 x - \operatorname{sech}^2 x)\, dx > \frac{4}{1+e^{\pi}}.$$

**Question 9.** Find:

a) $\displaystyle\int \sinh(2x+5)\,dx$

b) $\displaystyle\int 8\cosh(3-4x)\,dx$

c) $\displaystyle\int 2x\,\operatorname{sech}^2 x^2\,dx$

d) $\displaystyle\int x\sinh(3x^2-2)\,dx$

e) $\displaystyle\int \frac{\cosh\sqrt{x}}{2\sqrt{x}}\,dx$

f) $\displaystyle\int \frac{\operatorname{sech}^2(3/x^2)}{4x^3}\,dx$

**Question 10.** Evaluate the following definite integral:

$$\int_0^{\ln 2} \sinh\left(\frac{3x+\ln 4}{2}\right)\,dx$$

**Question 11.** Evaluate the following definite integral:

$$\int_0^{\sqrt{\ln 2}} 256x \cosh(4x^2)\, dx$$

**Question 12.** Show that:

$$\int_0^{\pi/2} \cos x \sinh(\sin x)\, dx = \frac{(e-1)^2}{2e}.$$

**Question 13.** Find:

a) $\int \tanh x \, dx$

b) $\int \coth x \, dx$

**Question 14.** Show that

$$\int_{\ln 2}^{\ln 4} (2\tanh 2x + 2\coth 2x)\, dx = \ln\left(\frac{p}{q}\right),$$

where $p$ and $q$ are integers and $\frac{p}{q}$ is in its simplest form.

**Question 15.**

a) Find $\dfrac{d}{dx}(\operatorname{cosech} x)$.

b) Hence find $\displaystyle\int \operatorname{cosech} x \coth x \, dx$.

**Question 16.**

a) Find $\dfrac{d}{dx}(\operatorname{sech} x)$.

b) Hence find $\displaystyle\int \operatorname{sech} x \tanh x \, dx$.

**Question 17.**

a) Find $\dfrac{d}{dx}(\coth x)$.

b) Hence find $\displaystyle\int \operatorname{cosech}^2 x\, dx$.

**The following questions in this section require the use of Integration by Substitution, Integration by Parts, Hyperbolic Double Angle Formulae and Partial Fractions.**

**Question 18.** Find:

a) $\displaystyle\int x \sinh 3x\, dx$

b) $\displaystyle\int x^2 \cosh \frac{x}{4}\, dx$

**Question 19.** The curve has gradient function

$$\frac{dy}{dx} = 9(x+2)\sinh 3x$$

and passes through the point $(0, 8)$. Show that $y$ can be written in the form

$$y = (ax+b)\cosh 3x - \sinh 3x + c,$$

where $a$, $b$ and $c$ are integers.

**Question 20.** Find:

a) $\displaystyle\int e^{x}\sinh x\,dx$

b) $\displaystyle\int e^{2x}\cosh x\,dx$

c) $\displaystyle\int e^{3x}\sinh 2x\,dx$

d) $\displaystyle\int e^{-4x}\cosh 4x\,dx$

**Question 21.** Find $\int \cosh^2 x \, dx$.

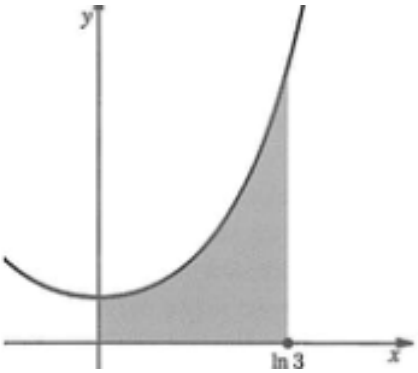
**Question 22.** Find $\int \sinh^2 6x \, dx$.

**Question 23.** Show that

$$\int_{\ln(5/6)}^{\ln(6/5)} \cosh^2 \frac{x}{2}\, dx = p + \ln q,$$

where $p$ and $q$ are positive rational numbers.

**Question 24.** The diagram below shows the curve $y = 4\cosh^2 x - 3\cosh x$. Show that the exact shaded area is $\frac{4}{9} + \ln 9$.



**Question 25.** Find:

a) $\int \sinh^3 x \, dx$

b) $\int \cosh^3 x \, dx$